

WESTERLY, RI, USA

WMO: 722151

Lat: 41.350N

Lon: 71.799W

Elev: 81

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 14794

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.5	(c) 13.4	(d) -8.9	(e) 3.4	(f) 11.5	(g) -4.4	(h) 4.4	(i) 18.0	(j) 23.8	(k) 34.7	(l) 20.8	(m) 36.9	(n) 6.8	(o) 310	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	85.6	72.8	83.1	71.8	81.0	70.7	76.4	81.9	74.8	79.8	73.4	77.7	8.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	130.0	79.3	73.2	123.5	77.4	72.0	118.6	76.3	39.7	81.8	38.2	80.0	37.0	78.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
19.0	16.9	15.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			0.6	90.8	6.4	2.9	-4.0	92.9	-7.7	94.6	-11.3	96.2	-16.0	98.4
			-0.7	78.1	5.9	1.8	-4.9	79.4	-8.4	80.5	-11.7	81.5	-16.0	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	30.8	31.7	38.0	47.9	57.2	65.8	72.7	71.4	65.1	54.8	44.7	36.7		
	DBStd	16.25	9.64	8.60	7.94	6.37	6.27	5.56	4.49	4.82	6.08	7.58	7.81	8.65		
	HDD50	2262	597	512	379	113	9	0	0	0	0	38	195	419		
	HDD65	5548	1062	931	838	515	253	55	2	5	74	326	610	877		
	CDD50	2813	1	1	6	49	233	475	704	663	452	187	35	7		
	CDD65	625	0	0	0	1	12	81	241	203	77	10	0	0		
	CDH74	3507	0	0	0	11	60	390	1477	1257	294	18	0	0		
	CDH80	672	0	0	0	2	6	63	330	236	32	3	0	0		
(14) Wind	WSAvg	6.7	7.3	7.4	7.9	7.7	6.8	6.2	5.8	5.5	5.8	6.6	6.8	6.9		
(15) Precipitation	PrecAvg	44.7	3.5	3.1	4.4	4.0	3.4	3.3	3.2	4.0	4.2	3.9	3.4	4.3		
	PrecMax	54.9	6.0	5.6	14.2	7.7	7.7	11.1	11.3	8.2	8.6	13.0	5.6	7.4		
	PrecMin	34.5	1.6	1.6	0.5	1.3	1.1	0.0	1.3	1.5	0.9	0.4	0.8	0.9		
	PrecStd	5.8	1.2	1.2	2.8	1.9	1.5	2.6	1.9	1.9	1.9	2.6	1.4	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	54.3	63.1	73.8	80.2	85.9	90.2	88.3	83.5	76.1	65.5	59.9		
		MCWB	52.4	48.3	52.6	57.8	66.3	71.0	75.6	74.8	72.6	66.9	59.6	56.2		
	2%	DB	51.7	50.8	56.4	65.9	74.9	81.7	86.3	84.9	79.9	71.8	62.9	56.8		
		MCWB	50.2	47.1	47.4	52.2	63.5	69.2	73.3	73.2	69.3	66.7	59.0	54.3		
	5%	DB	48.3	48.0	52.6	61.4	71.3	78.7	83.7	82.6	77.2	69.6	60.6	54.1		
		MCWB	45.3	44.2	46.7	51.4	61.0	67.3	72.3	72.0	68.9	65.0	56.4	51.8		
	10%	DB	45.5	44.9	49.6	58.1	67.9	75.5	81.3	80.5	75.0	67.0	58.2	51.0		
		MCWB	42.6	41.0	44.3	49.9	59.1	65.7	71.3	71.1	68.4	62.1	54.4	47.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	51.9	55.4	60.7	69.0	74.0	78.5	78.2	75.7	71.0	63.7	58.2		
		MCDB	54.1	53.0	58.2	66.9	76.7	82.3	86.7	84.3	79.4	72.6	64.6	59.2		
	2%	WB	50.6	48.2	51.1	57.1	66.1	71.5	76.2	76.8	73.6	68.4	60.3	54.9		
		MCDB	51.3	49.5	54.0	61.7	72.3	78.5	82.8	81.7	76.9	70.7	61.8	56.1		
	5%	WB	46.3	45.1	48.0	54.6	63.7	69.8	74.8	75.1	71.8	66.0	57.8	52.3		
		MCDB	47.7	47.7	51.3	58.4	68.9	75.3	80.2	79.7	74.8	68.7	59.5	53.7		
	10%	WB	42.7	41.6	45.3	52.0	61.4	68.2	73.4	73.5	70.1	63.6	55.2	48.3		
		MCDB	45.1	44.4	48.6	55.6	65.8	72.7	78.0	77.4	73.3	66.1	57.9	50.7		
(35) Mean Daily Temperature Range	MDBR		14.7	15.2	15.0	16.2	15.5	15.0	14.7	15.7	16.5	16.5	16.2	14.2		
	5% DB	MCDBR	17.0	17.0	18.3	21.4	19.8	18.8	17.4	16.9	17.1	17.0	16.7	16.1		
		MCWBR	17.0	14.4	12.6	11.6	10.3	9.1	7.6	8.1	10.2	12.0	15.0	16.4		
	5% WB	MCDBR	17.0	17.4	15.8	16.3	16.4	15.5	14.1	13.0	12.9	13.8	14.5	15.8		
		MCWBR	17.8	16.6	12.9	11.4	10.0	8.8	7.1	7.2	9.0	11.3	14.9	17.0		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.315	0.338	0.375	0.423	0.464	0.488	0.450	0.379	0.345	0.317	0.304		
	taud		2.495	2.471	2.449	2.378	2.276	2.202	2.145	2.243	2.414	2.516	2.553	2.553		
	Ebn at Noon		270	285	290	284	272	260	252	258	272	270	265	262		
	Edh at Noon		23	28	32	37	42	45	48	42	33	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		592	872	1197	1550	1720	1882	1920	1723	1408	951	651	496		
	RadStd		48	84	90	166	167	188	103	95	107	88	49	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.15	N/A	N/A	+2.43	N/A	N/A	N/A	N/A	+353	+184
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	+99

Nomenclature: See separate page